

Nanoscale Device Design: A TCAD Analysis of 2nm GAAFET Nanosheet

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Abstract

The size of field effect transistors used in CMOS integrated circuits shrinks over each technology node to achieve lower power consumption, higher performance, and higher density. The increase in electrostatic control and reduced short channel effects (SCE) are key benefits to adopting new architectures like Gate-All-Around FET (GAAFET) to meet scaling requirements for next generation process nodes. Advanced architectures in the nanometer regime bring unique design challenges that require thorough characterization of device performance to ensure successful and robust implementation. Sentaurus TCAD provides a valuable tool to begin exploring the many design choices faced when encountering a novel structure such as GAAFET. Design choices around device dimensions, geometries, and materials can be simulated to understand impacts on performance. In this work, several of these design choices are explored, such as channel geometry, vertical channel stacking, and spacer material options, all in the context of projected parameters for the 2nm technology node. Key metrics like transfer characteristics, SCE susceptibility, extension resistance, and gate capacitance are used to create a basis for evaluating the device design robustness. The same n-type and p-type devices are then simulated using TCAD mixed mode circuit simulation to further explore device performance and evaluate tradeoffs.

Acknowledgments

I would like to thank my advisor Dr. Feng Li for the guidance and advice through this process.

Dedication

To Mylee, for so many amazing years and unconditional support.

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Chapter 1: Introduction

Power, Performance, Area, and Cost (PPAC) demand smaller devices at each technology node. Reduced operating voltages necessitate smaller channel lengths to achieve higher speed and performance node after node. For decades the transistor architecture has remained largely the same, constructed in a 2-dimensional plane with a single gate interface. Significant engineering resources contributed to the continued device scaling for the planar architecture to counteract worsening short channel effects (SCE) and improve overall performance. Some of these key contributions include high-k metal gate, strain engineering, ultra-thin body silicon on insulator (UTB-SOI), and a mixture of Silicon with novel materials (e.g. Germanium). The introduction of the FinFET was the first major industry adopted architectural change to create a 3D structure with multi-gate interface. The FinFET extends the channel in the z-axis to form a tall “fin”, with gate interface on three sides of the channel. The primary driver for this paradigm shift is the increasingly detrimental SCE and excessive leakage current that make planar devices infeasible at the advanced technology nodes. Adopting a multi-gate architecture significantly increases the electrostatic control over the channel and the subsequent increase in effective width (W_{eff}) helps to boost current density without increased footprint. FinFET has delivered promising results in both SCE suppression and overall performance for the 10nm, 7nm, and 5nm nodes. However, regarding 3nm and 2nm node, it is expected that even three gates will no longer be able to sufficiently suppress SCE and industry will naturally progress to four gates that fully encompass the channel on all sides (i.e. Gate-All-Around FET, GAAFET). The GAAFET structure typically takes the form of either a nanowire (NW) or nanosheet (NS). For NW, the channel is a cylindrical or square shape, while NS (also referred to as MBCFET or RibbonFET) is typically a thin rectangular shape. The width of the device can be modulated to provide designers flexibility in tuning performance for a given application. Additionally, GAAFET enables the ability to stack channels vertically, one over the other, to increase drive current without sacrificing larger dimensions in the x and y axes. Figure 1.1 illustrates the geometry differences between planar, FinFET, and NS (i.e., MBCFET) devices.

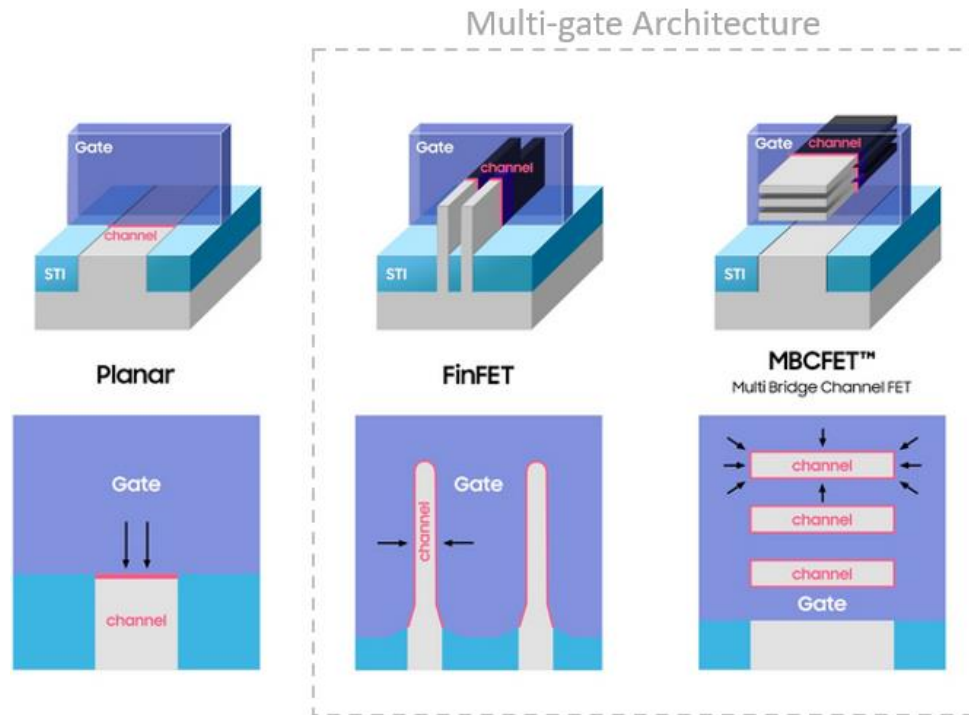


Figure 1.1 Planar, FinFET, and GAAFET (i.e., MBCFET) [“3NM GAA MBCFET: A Cutting-Edge Advancement That Brings Unrivaled SRAM Design Flexibility”].

Device Scaling

Transistor scaling can hardly be mentioned without first acknowledging Moore’s Law, which has accurately predicted semiconductor scaling trends for the past several decades. This law is more of an economic observation, made by Gordon Moore (co-founder of Intel) in the 1960’s, that stated the number of transistors in an integrated circuit will double approximately every two years. Dennard scaling of MOSFET devices has enabled this prediction by requiring transistor area to scale by a factor of 0.5x, meaning a 0.7x reduction in both the x and y plane (i.e., $\sqrt{0.7} \approx 0.5$), illustrated by Figure 1.2. As the device area shrinks, so does chip capacitance and the necessary voltage and current to maintain operation. The electric field (V/m) will remain constant when the distance and voltage decrease accordingly, which allows for the increase in operating frequency while overall power density remains constant.

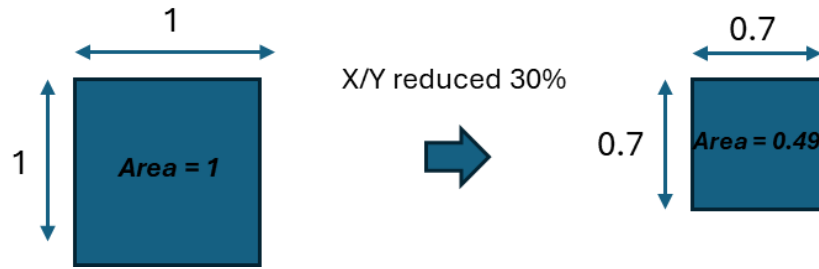


Figure 1.2 Dennard scaling of semiconductor chip area.

In the past two decades this scaling trend has been increasingly difficult to maintain. Dennard scaling in the traditional sense reached its limit due to increased leakage and parasitic impacts. This trend results in the gradual tapering of gate length scaling in the mid 2000's shown by Figure 1.3. The nomenclature around channel length and technology node diverged several years before this, around the 350nm node. At this point, the channel length was scaled aggressively at a rate greater than 0.7x, however, the technology node maintained this interval with more relevance to metal half-pitch. With the introduction of the FinFET, the technology node name effectively became meaningless to any specific device dimension.

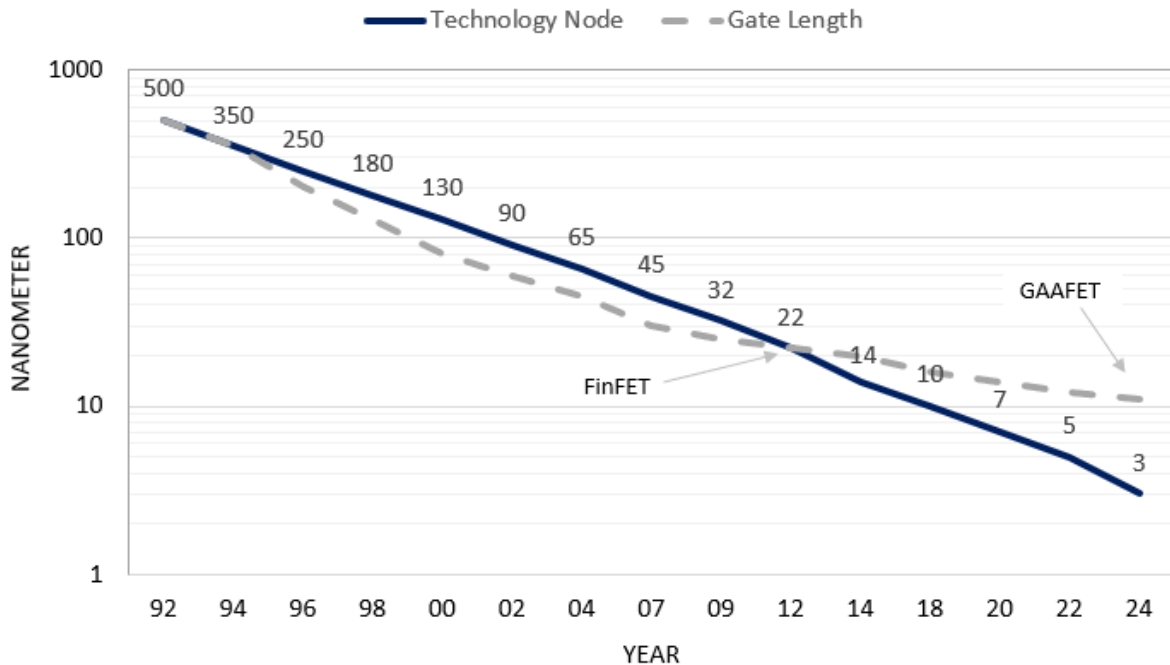


Figure 1.3 Semiconductor technology node versus gate length [Yu, 2022].

Short Channel Effects (SCE)

The transition from gate lengths measured in micrometers (i.e., microelectronics) to state-of-the-art channel lengths in the tens of nanometers (i.e., nanoelectronics) has brought with it an abundance of technical challenges. A standard MOSFET log scale I_{ds} versus V_{gs} curve is shown in Figure 1.4.

Several primary characteristics of this curve are illustrated, including I_{on} current, I_{off} current, drain induced barrier lower (DIBL), subthreshold swing (SS), and threshold voltage (V_t). The I_{on} current is referring to the I_{ds} value when $V_{gs}=V_{dd}$. Similarly, I_{off} current is the I_{ds} value when $V_{gs}=0V$. The degradation of certain characteristics as channel lengths decrease has coined the name “short channel effects”. Some of the most common concerns result from a more gradual subthreshold slope, larger DIBL shift, decreased V_t , and large I_{off} .

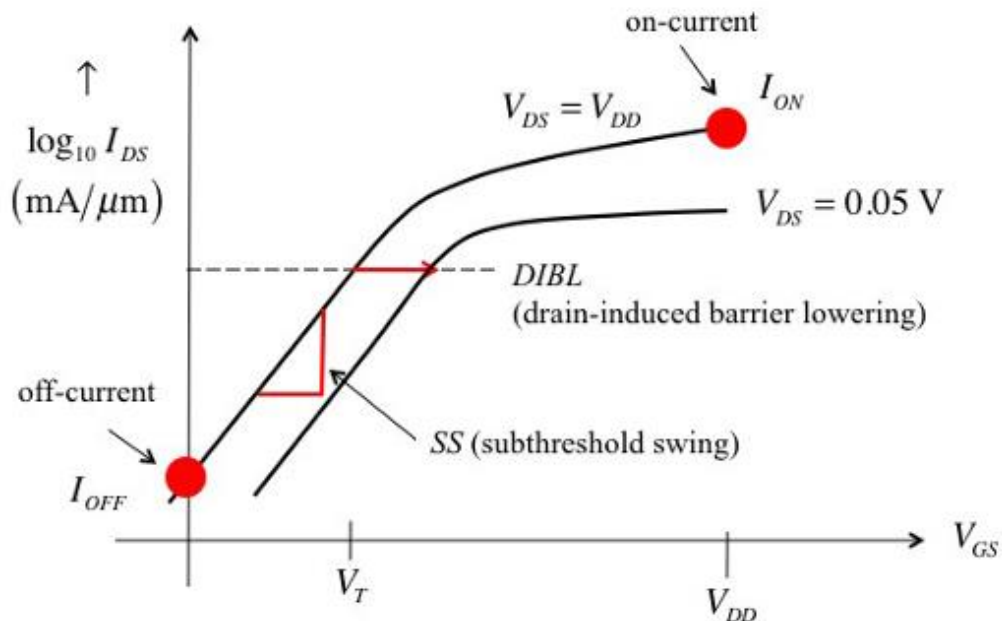


Figure 1.4 nMOS transfer characteristics [Lundstrom, 2018].

MOSFET Operation

Traditional MOSFET physics states that current is proportional to the width of the device, the amount of charge in the channel, and the carrier velocity. Current flow from source to drain is controlled by applying voltage to the gate to increase or decrease the energy barrier between source and drain. The shape of the I-V curve is a function of device electrostatics from charge in the channel, while the magnitude of the current is dependent on the transport (i.e., velocity). Electrostatics and transport are the two primary components of transistor physics.

Electrostatics

A positive electrostatic potential lowers the energy of an electron. This is what enables the gate to control the energy barrier present in the channel. Poisson's equation describes the electric potential of the channel in three-dimensional space. Energy band diagrams provide a convenient qualitative solution to Poisson's equation. When a negative gate voltage is applied, both the conduction and valence energy bands bend upward. For an n-type device, this has the effect of accumulating holes as the valence band bends toward the Fermi level. When applying a positive voltage, the bands bend down, which depletes the channel of holes. If the conduction band is bent far enough, the conduction band bends toward the Fermi level and the channel inverts as free electrons gather under the gate. The band diagrams for accumulation, flat band, and depletion/inversion regimes of operation are illustrated in Figure 1.5. For small channel devices, the surface potential and charge in the channel is a function of both V_{gs} and V_{ds} . Figure 1.6 illustrates a 'well behaved' energy band structure along the channel length in the presence of V_{ds} applied across source and drain terminals.

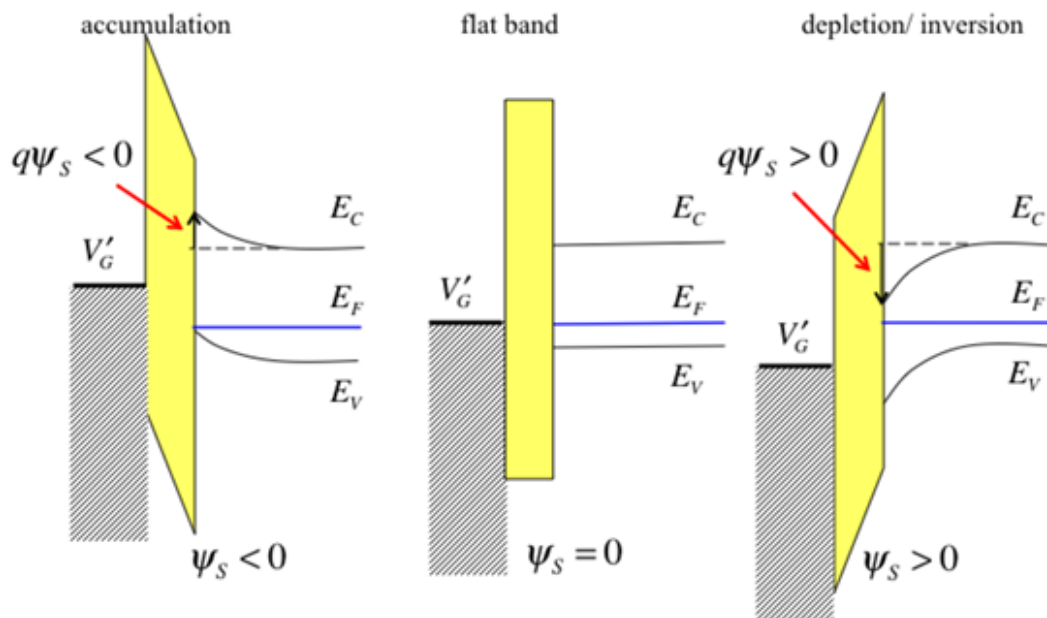


Figure 1.5 Energy band diagrams illustrating the band bending for accumulation, flat band, and depletion/inversion regions of operation [Lundstrom, 2018].

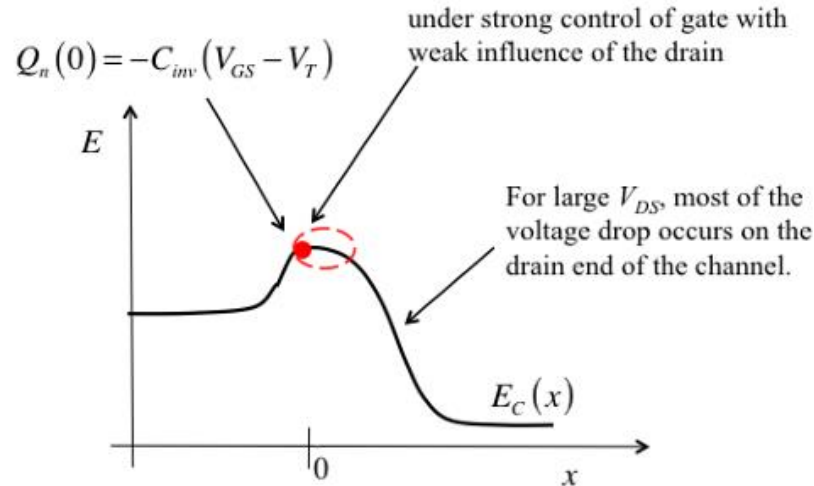


Figure 1.6 Energy band along the length of the channel showing band bending due to drain voltage for device with strong gate control [Lundstrom, 2018].

Transport

The carrier velocity (i.e., drift velocity) is proportional to the electric field as given by $V_d = \mu E$. Electron and hole velocity are often different, even under the same electric field. Semi-classical transport using Boltzmann Transport Equations (BTE) and Drift Diffusion (DD) has remained the basis for carrier transport modeling in transistors for many decades. DD is derived from the BTE equations and works well to describe long channel devices ($L_g > 1\mu\text{m}$). For short channel devices, gate lengths between 10nm and 100nm, a more accurate representation of transport is often needed. Without transitioning to more computationally expensive transport modeling, a quantum confinement correction term can be included with the DD to account for the quantum confinement that occurs at the channel / gate oxide barrier. The reduction of free carriers affects the current. This type of correction allows DD modeling to be used into the deca-nanometer regime. However, as channel lengths shrink even further (e.g., $L_g \leq 10\text{nm}$) carrier transport approaches the ballistic regime where the velocity no longer saturates (i.e., minimal scattering occurs). This quasi-ballistic regime occurs when the mean-free time (i.e., the time between a carrier colliding with other carriers, impurities, phonons, etc.) is approximately the same length as the channel. To accurately model carrier movement requires using the carrier wave function described by Schrödinger's time-dependent equation.

FinFET

The change to a multi-gate, FinFET architecture became a necessary tool to overcome SCEs in the most recent advanced technology nodes. A standard FinFET cell typically employs multiple fins for a single device to meet the required current density while abiding by end application leakage constraints. Standard cell scaling has become increasingly difficult due to cell size limitation by fin-to-fin pitch, and minimum fin to source/drain (S/D) width ratio to maintain acceptable resistance. Due to the limitations, scaling requires a discrete reduction in terms of the number of fins per cell. To maintain current drive with fewer fins, the fin heights are extended to increase W_{eff} , however, this can lead to process challenges reliably meeting higher aspect ratios. Figure 1.7 depicts the fin pitch and height scaling between technology nodes to improve performance and meet scaling requirements. Along with logistical challenges, the difficulties to maintain the electrical effectiveness that garnered the mass adoption of the FinFET in the first place has become increasingly difficult to manage.

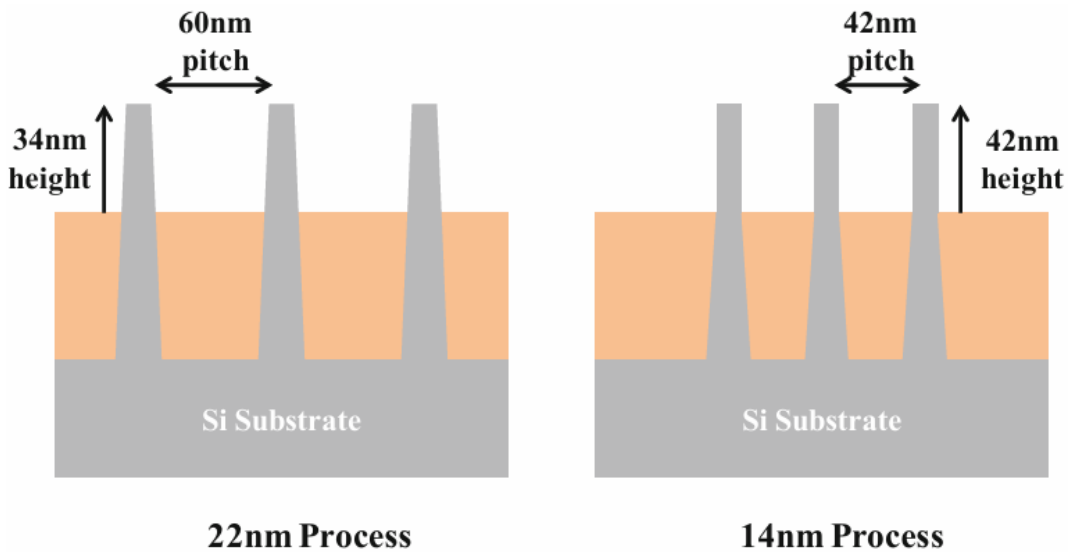


Figure 1.7 Differences of Fin pitches and heights for 22nm and 14nm FinFET [Wu, 2018].

Gate-All-Around FET (GAAFET)

The transition to GAAFET architecture strives to solve many of the shortcomings faced by FinFET. Several different types of GAAFETs exist depending on the orientation and shape of the channel. Broadly, these can be categorized as either lateral or vertical, and NW or NS based on the channel dimensions and shape. Although a lot of research is conducted on all types of GAAFET, the most promising based on industry trends appears to be lateral NS GAAFET. Like other MOSFET architectures, the substrate of the device can consist of either bulk silicon or buried oxide (BOX).

Bulk silicon has continued to be the mainstream option due to cheaper implementation but has the drawback of introducing a parasitic channel under the primary channels, that resembles a planar MOSFET due to the single gate interface. Several tactics have been explored to reduce the impact of this undesirable channel, such as modulating the parasitic channel shape to match more closely to the primary channels [Yunho, et al, 2020] and high doping leakage suppression [Jegadheesan, et al, 2019]. The higher doping concentration increases the threshold voltage to offset from the primary channels, thereby reducing the leakage current contributed. An alternative option is to use buried oxide approach that forms a thin layer of SiO₂ under the GAAFET structure and eliminates the parasitic channel all together.

Chapter 2: GAAFET Device Simulation

Sentaurus TCAD is used for the device and circuit characterization of GAAFET NS structures [Sentaurus TCAD, 2022]. The device is generated using Sentaurus Structure Editor (SDE), which defines the materials, doping profile, and structure dimensions. Sentaurus Device (SDEVICE) is then used to define gate electrical characteristics, work-function, and physics models for device simulation. Many dimensional parameters are based on IRDS projected dimensions of the 2nm node, summarized in table 2.1 [IRDS, 2022].

Table 2.1 Device parameter setup for TCAD simulation

N2 Parameter Setup for GAA	nMOS	pMOS
Contacted Gate Pitch (CGP) (nm)	45	45
Gate Length (L_g) (nm)	14	14
Effective Oxide Thickness (EOT) (nm)	1.32	1.32
Spacer (nm)	6	6
Contact (nm)	19	19
Stack Spacer (nm)	20	20
Channel Doping (cm^{-3})	2.E+17	2.E+17
Source/Drain Doping (cm^{-3})	4.E+20	4.E+20
Work function (eV)	4.52	4.77

Figure 2.1a and 2.1b show a single channel and dual channel device, while Figure 2.1c and 2.1d show the cross section along the Y-Z axis of the channel, respectively. Figure 2.2a and 2.2b show the X-Z cross section, highlighting the doping concentration profiles of the S/D and channel regions. The different cross section views enable easier visualization of the many material layers and region interfaces for these complex 3-dimensional structures.

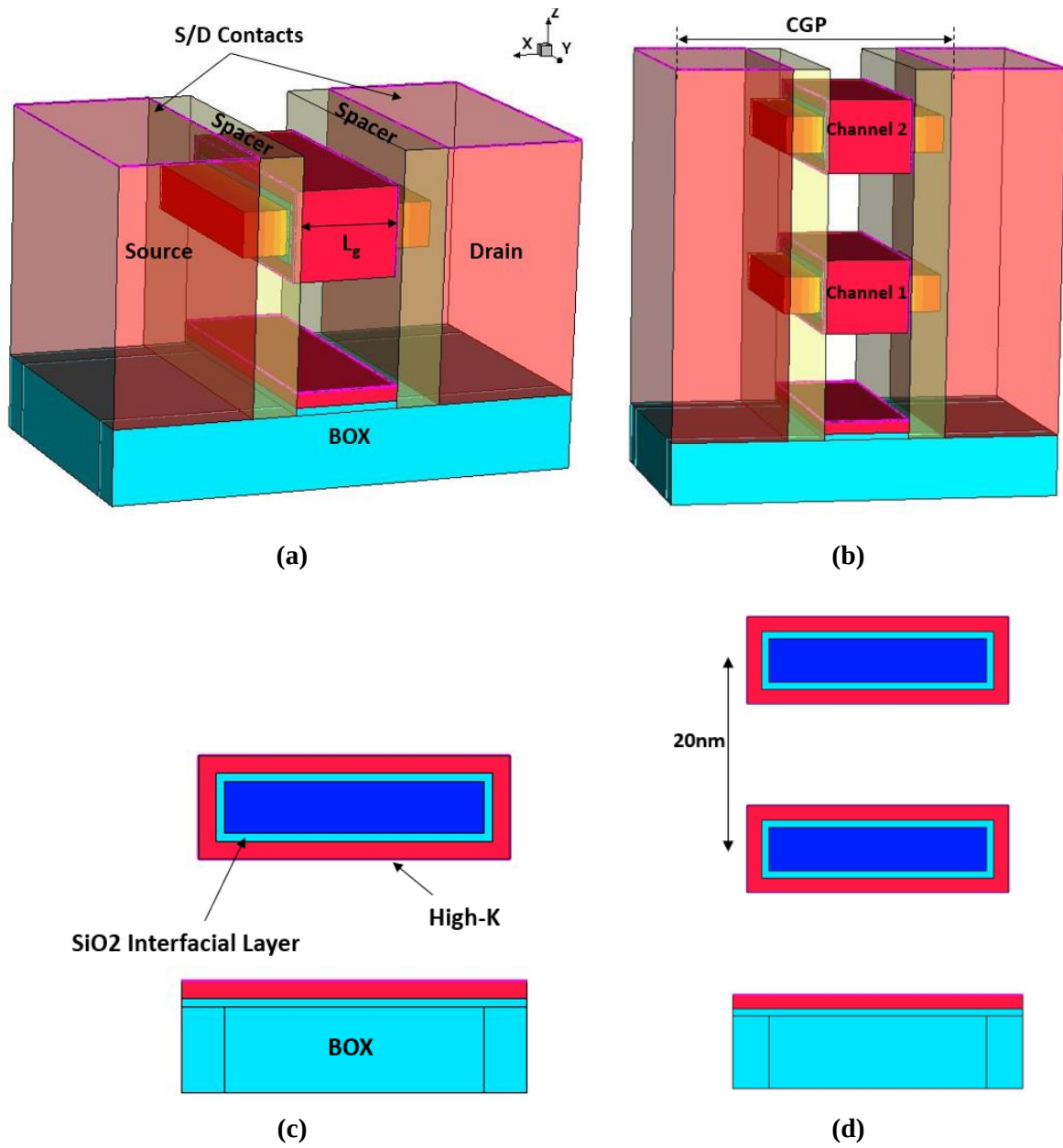


Figure 2.1 GAAFET SDE Model for (a) single and (b) dual channel devices, with channel cross section (Y-Z direction) in (c) and (d) respectively.

Several physics models are used conjointly to describe different aspects of material interfaces and device operation. A thin layer mobility model is incorporated to accurately describe channel geometry dependencies like electrical resistance due to surface roughness and scattering mechanisms. Using high-k stack can introduce remote scattering effects from the SiO₂ and HfO₂ interface. The Density Gradient (DG) quantum confinement correction term is included for the DD model to

a consequence of the reduced energy barrier, and a very prominent increase in I_{off} current due to GIDL as V_{gs} decreases negatively.

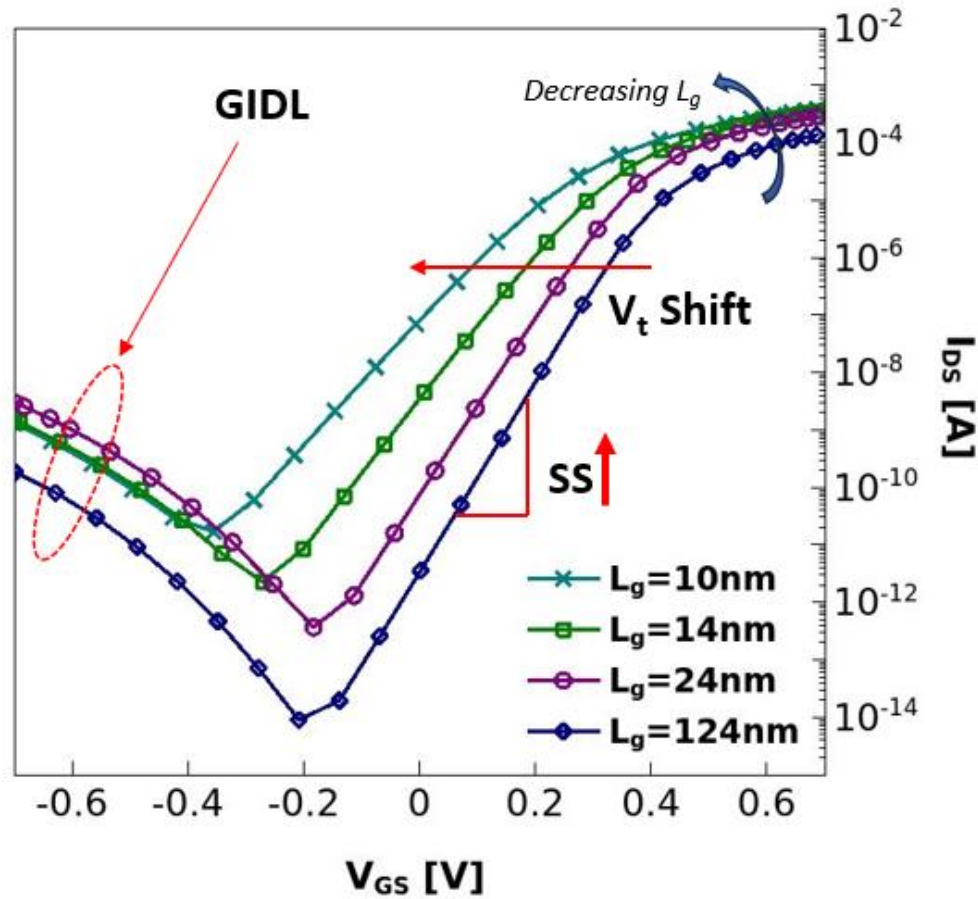


Figure 2.3 Gate voltage versus drain current for gate lengths 10, 14, 24, and 124nm.

Drain Induced Barrier Lowering (DIBL)

The potential barrier that separates the source and drain begins to see reduced height as these terminals become more electrostatically coupled. The voltage applied to the drain has increased influence on the barrier height the shorter the channel gets. This phenomenon is problematic since the potential barrier becomes a function of both V_{gs} and V_{ds} and has the effect of the drain voltage moving the threshold voltage (V_t) level of the device. Ultimately, the device performance changes depending on what drain voltage is applied. This effect can be measured by first generating an $I_{\text{ds}}-V_{\text{gs}}$ curve in the linear mode of operation ($V_{\text{ds}} < V_{\text{gs}} - V_t$), then again in the saturation region ($V_{\text{ds}} > V_{\text{gs}} - V_t$), and using Equation 2.1.

$$DIBL = \frac{V_{t_{lin}} - V_{t_{sat}}}{V_{ds_{sat}} - V_{ds_{lin}}} \quad (2.1)$$

The conduction band energy diagram in Figure 2.4 shows a clear trend in DIBL impact on barrier height. The quasi-Fermi level of the drain is lowered by factor of qV_{ds} , which distorts equilibrium and means that the device will begin conducting current for carriers with enough energy to overcome the resulting barrier. The lower the barrier becomes, the less energy is required.

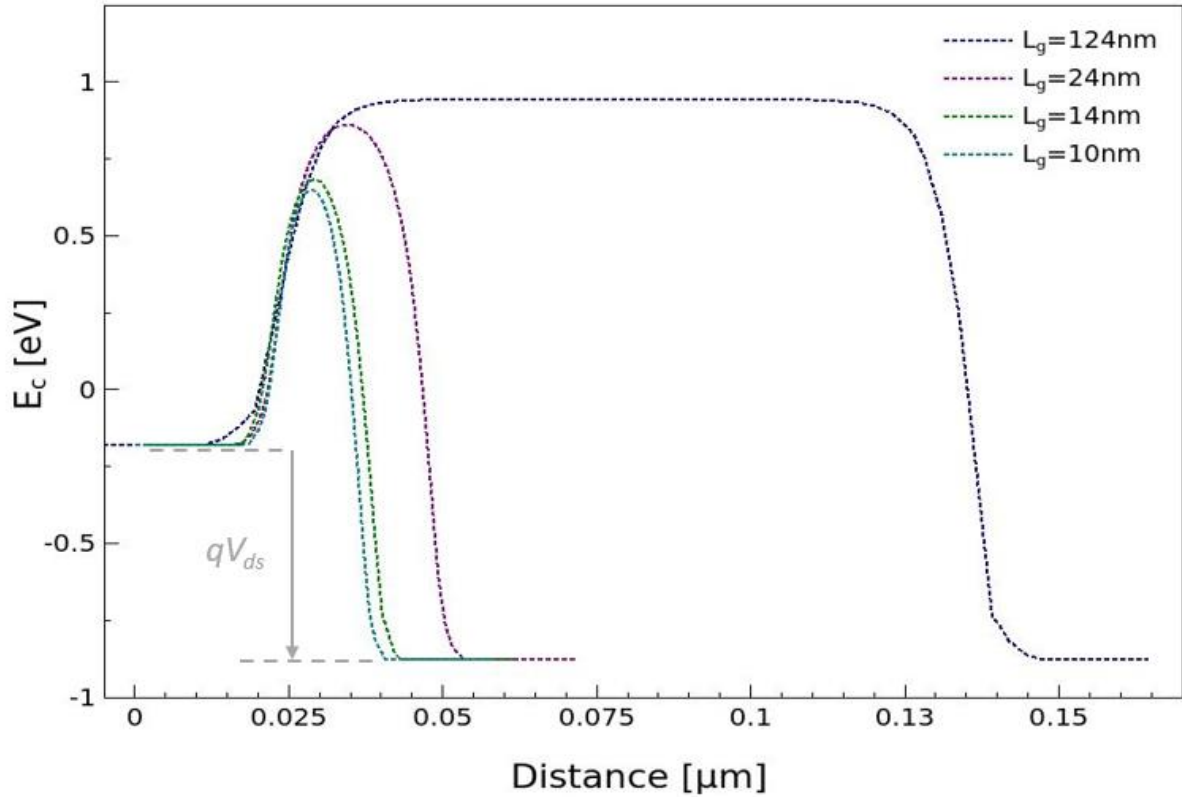


Figure 2.4 Energy band diagram (along channel length) versus device gate length.

Subthreshold Swing (SS)

The logarithmic decrease in current below the threshold voltage is an important metric to consider when characterizing a device. This portion, known as the subthreshold swing, is present for all devices and is not an SCE by itself, however, the degradation due to channel length scaling is. This metric provides insight to the level of current that will continue to pass through a device even when it is off (I_{off}). This “leakage” current is typically small for a single device (e.g., nA to pA range), however, multiplied by billions of devices, this becomes a major concern for VLSI design, especially

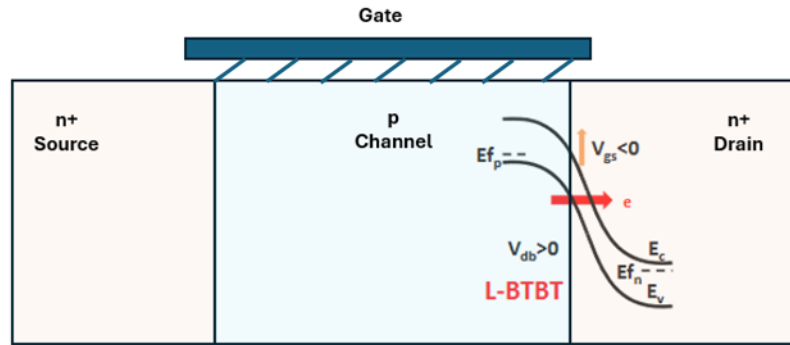
in portable electronic applications. A lower SS will allow for a lower V_t resulting in higher I_{on} current while maintaining the same I_{off} current. SS is given in Equation 2.2,

$$SS = \frac{\partial V_{gs}}{\partial \ln I_d} = \frac{\partial V_{gs}}{\partial \phi_s} \frac{\partial \phi_s}{\partial \ln I_d} = \frac{kT}{q} \ln 10 \left(1 + \frac{C_{dep}}{C_{ox}} \right) \quad (2.2)$$

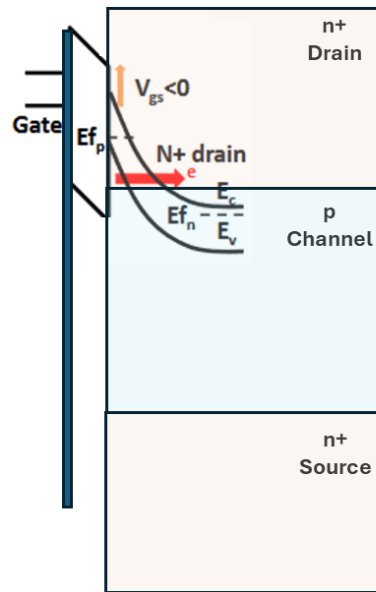
where kT/q is thermal voltage, C_{dep} is depletion capacitance, and C_{ox} is gate capacitance. The $\partial V_{gs}/\partial \phi_s$ describes the gate to channel coupling, referred to as the body factor. The relationship between gate voltage (V_{gs}) and surface potential (ϕ_s) is governed by the capacitive voltage divider between C_{ox} and C_{dep} . If C_{dep} is very small or C_{ox} is large (i.e. strong gate to channel coupling), SS will approach the minimum limit of 60 mV/dec. The inverse of SS gives the slope of the subthreshold curve.

Gate Induced Drain Leakage (GIDL)

An important, but easily neglected, region of the device I-V characteristic curve is the range where V_{gs} becomes negative. In this range, the device can experience elevated GIDL due to severe conduction and valence band bending. The components of GIDL current typically arise from BTBT and can usually be categorized into two main pieces, lateral BTBT (L-BTBT) and transversal BTBT (T-BTBT). L-BTBT describes the lateral tunneling that occurs at the pn junction between the channel and the drain that usually dominates at smaller negative V_{gs} voltages. The diagram in Figure 2.5a illustrates the band bending that can be aggravated by increasingly negative V_{gs} . Steep doping profile transitions can exacerbate this mechanism. As V_{gs} decrease more negatively, T-BTBT can become more dominate if there is significant overlap between gate and drain regions as exaggerated in Figure 2.5b. Extremely negative V_{gs} can lead to direct tunneling through the gate oxide.



(a)

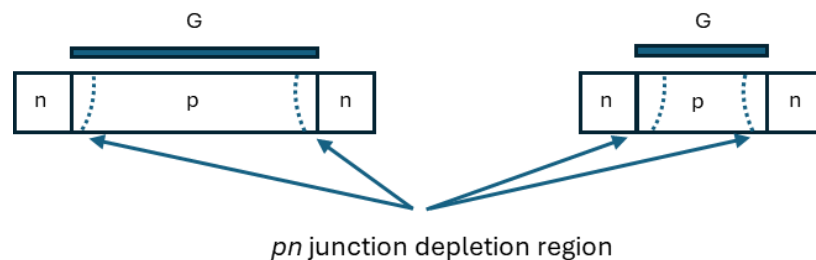


(b)

Figure 2.5 Band to band tunneling in (a) lateral and (b) transversal direction [J. Fan *et al*, 2013].

V_t Rolloff

For an n-type MOSFET to begin conducting (i.e., “turn on”), the gate voltage is increased which first depletes the channel of holes to a depth W_{dm} . As gate voltage continues to increase, an inversion layer of electrons is created forming a channel. At this point, the device is “on” and current will flow with applied voltage across the S/D contacts. For short channel devices, the depletion regions formed at the S/D *pn* junctions account for a larger contribution to the overall channel depletion compared to long channel, depicted by Figure 2.6. Therefore, lower gate voltage is required to deplete the channel to a depth, W_{dm} , and form the inversion layer. From an energy band diagram perspective, this effect can result in a lower energy barrier height.

Figure 2.6 *pn* junction depletion region influence on v_t Rolloff.

Parasitic Elements

Inherent to all MOSFET devices are the parasitic elements of resistance and capacitance that come with design and architecture choices. At the advanced nodes, devices have moved to 3D architectures and with it comes increased parasitic components. These parasitic values begin to approach similar magnitudes of the device parameters and if not properly managed, can severely limit the performance of the device.

Resistance

Series resistance contributed from the S/D regions will limit overall I_{on} capability of the device. The total resistance associated with a GAAFET device can be broken down into two main categories as shown in Equation 2.3,

$$R_{on} = R_{ch} + R_{SD} \quad (2.3)$$

where R_{on} is the total resistance of the device in linear mode of operation, R_{ch} is the resistance contributed from the channel, and R_{SD} is the resistance contributed from the S/D extensions. To estimate the contribution of each of these parts, an R_{on} versus gate length analysis is used. R_{on} is plotted over a range of gate lengths and the results are extrapolated to provide a y-intercept that gives an approximate resistance for an effective gate length (L_{eff}) of zero. The source and drain are assumed to contribute one half of the resulting resistance for $L_{eff} = 0\text{nm}$. Figure 2.7 demonstrates this technique and provides the approximation for S/D resistance at the 14nm gate length, equal to 5.6K Ω . The channel resistance is then calculated using Equation 2.3, which equates to 1.3K Ω .

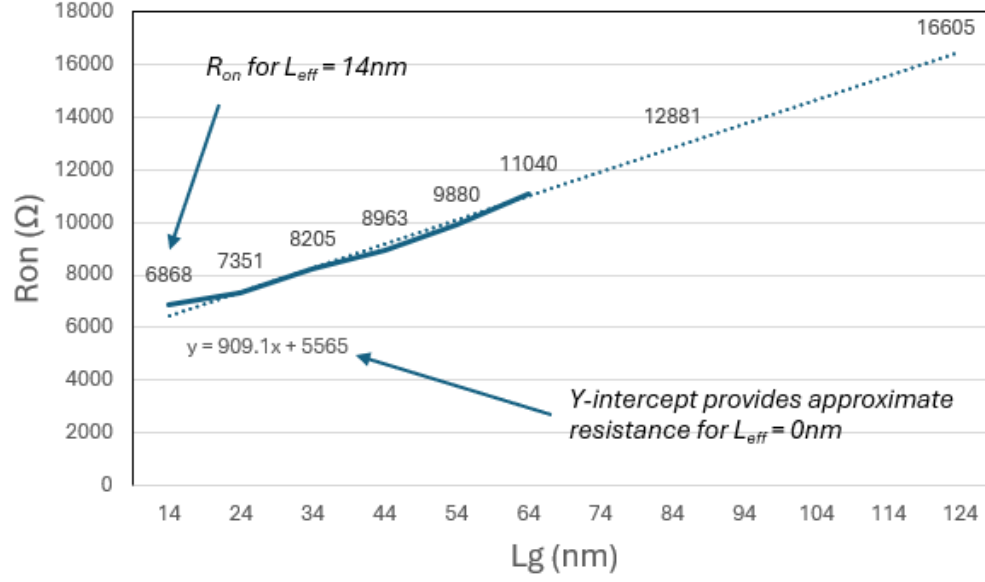


Figure 2.7 S/D resistance extraction.

The R_{SD} portion can be further broken down into subcategories that include the contact resistance, S/D spreading resistance, and spacer extension resistance using a conductance integration method if more granular categorization is needed [Sudarshan *et al*, 2016].

Gate Capacitance

CMOS technology power consumption can typically be categorized as either static or dynamic. The dynamic power draw is dominate and comes from the device switching logic states (i.e., logic zero to one or one to zero). The dynamic power draw can be calculated using Equation 2.4,

$$P_d = CfV_{dd}^2 \quad (2.4)$$

where C is the load capacitance, f the operating frequency, and V_{dd} the operating voltage. On the circuit level, the load capacitance equates to the gate capacitance of the circuit that is being driven. This gate capacitance is shown as a function of V_{gs} in Figure 2.8.

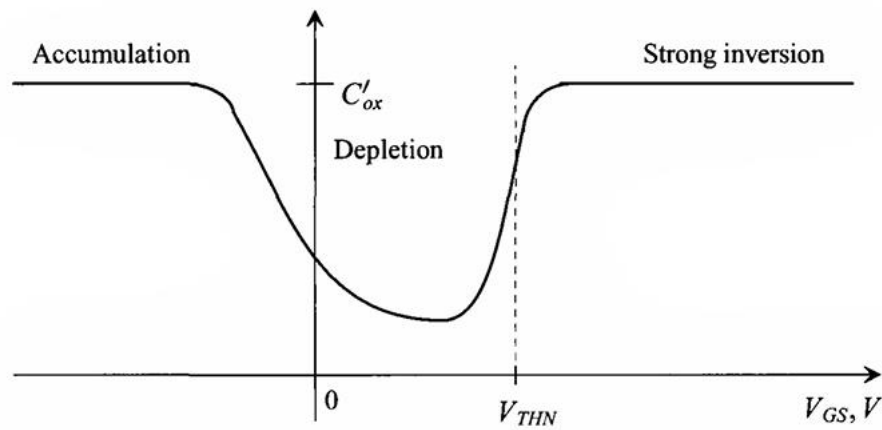


Figure 2.8 CMOS gate capacitance [Baker, 2019].

Along with power considerations, gate capacitance is also an important consideration for circuit speed. Large gate capacitance will increase switching delay and result in slower logic operations. Smaller channel widths tend to reduce parasitic capacitance between the S/D regions and if multiple channels are vertically stacked. The gate capacitance can be evaluated using TCAD small signal AC analysis options, where a complex matrix is calculated from resulting current changes to small voltage perturbation at a specified frequency. The imaginary part of this complex matrix is the result of the out of phase contribution from device capacitance. Figure 2.9 demonstrates gate capacitance for a single channel GAAFET as a function of V_{gs} , across different MOSFET regions of operation.

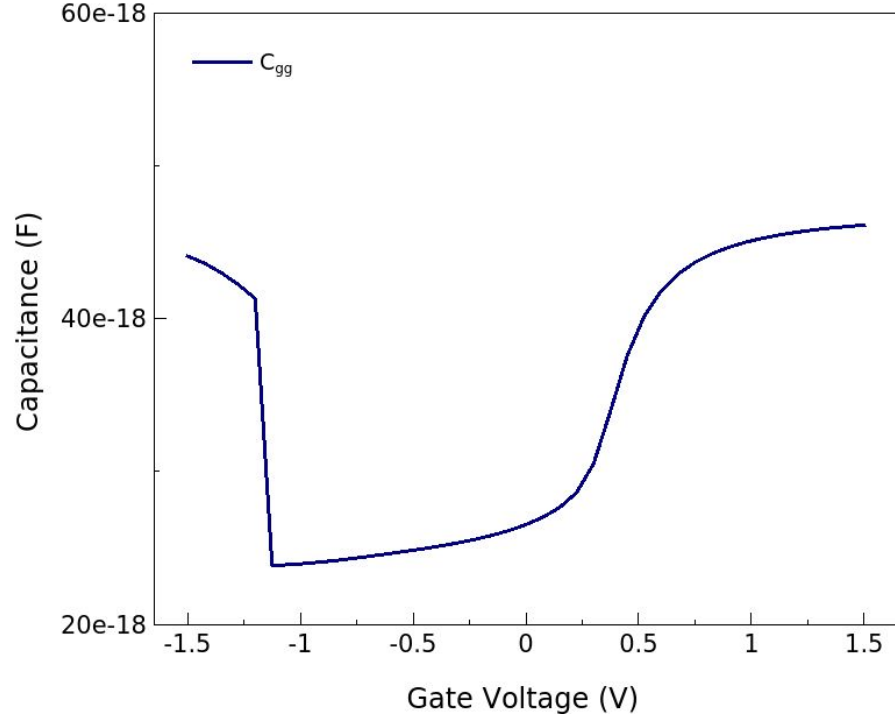


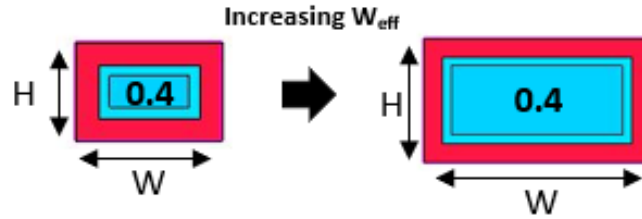
Figure 2.9 Gate voltage versus gate capacitance (C_{gg}) for single channel device.

Channel W_{eff}

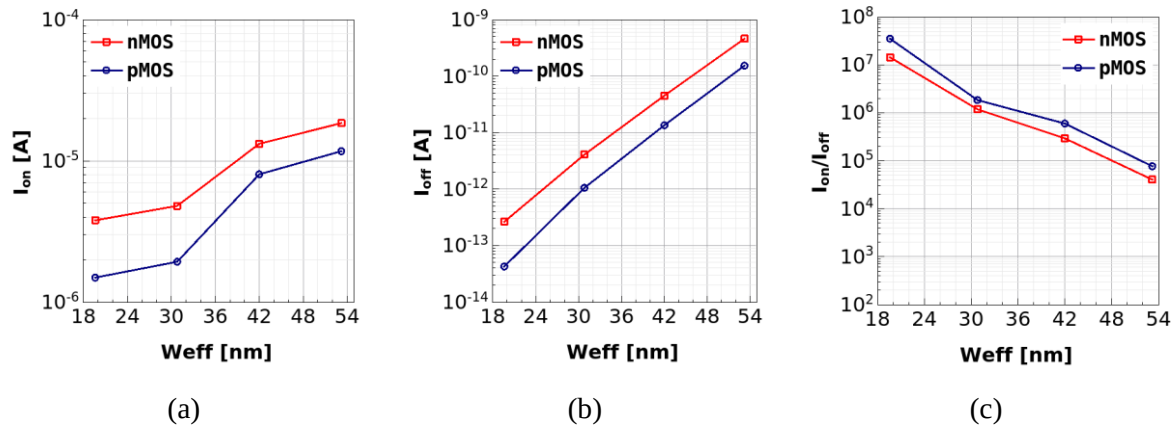
The effective width of the NS device is described in Equation 2.5,

$$W_{eff} = 2 \times (Width + Height) \times num_channels \quad (2.5)$$

where the perimeter of the NS device is calculated and multiplied by the number of vertically stacked channels (in this case $num_channels = 1$). To understand channel geometry impacts on performance, the channel effective width is adjusted around projected parameters of the 2nm node. Height and width parameters for low power devices are projected to be 6nm and 15nm, respectively. This analysis simply adjusts the height and width dimensions to change W_{eff} , while maintaining the projected height to width ratio of 0.4. Figure 2.10 illustrates the effect of increasing W_{eff} around this ratio.

Figure 2.10 W_{eff} analysis.

The results for I_{on} , I_{off} , and $I_{\text{on}}/I_{\text{off}}$ ratio are shown in Figure 2.11. Intuitively, a larger channel created with large W_{eff} has a significant increase in I_{on} current shown by Figure 2.11a. An apparent drawback to this is that the I_{off} current, depicted in Figure 2.11b, appears to increase at an even greater rate. Comparing $I_{\text{on}}/I_{\text{off}}$ ratio, this asymmetry is even more stark, shown by Figure 2.11c.

Figure 2.11 W_{eff} analysis for (a) I_{on} , (b) I_{off} , and (c) $I_{\text{on}}/I_{\text{off}}$ ratio

Examining Figure 2.11a results, there appears to be a significant increase in I_{on} current for channel W_{eff} larger than 30nm. The smaller channel dimensions will suffer from increased resistance due to scattering and quantum confinement effects, that will reduce overall I_{on} current. The extracted R_{on} values are plotted along with the I_{ds} versus V_{ds} curve in Figure 2.12. Significantly larger R_{on} is observed at the smaller W_{eff} values, that then steadily decrease for the larger W_{eff} values. This agrees with the I_{on} versus W_{eff} results shown in Figure 2.11a. A notable decrease in electrostatic control over the channel can be observed as the channel dimensions increase shown in subthreshold swing and DIBL calculations, Figure 2.13a and 2.13b, respectively.

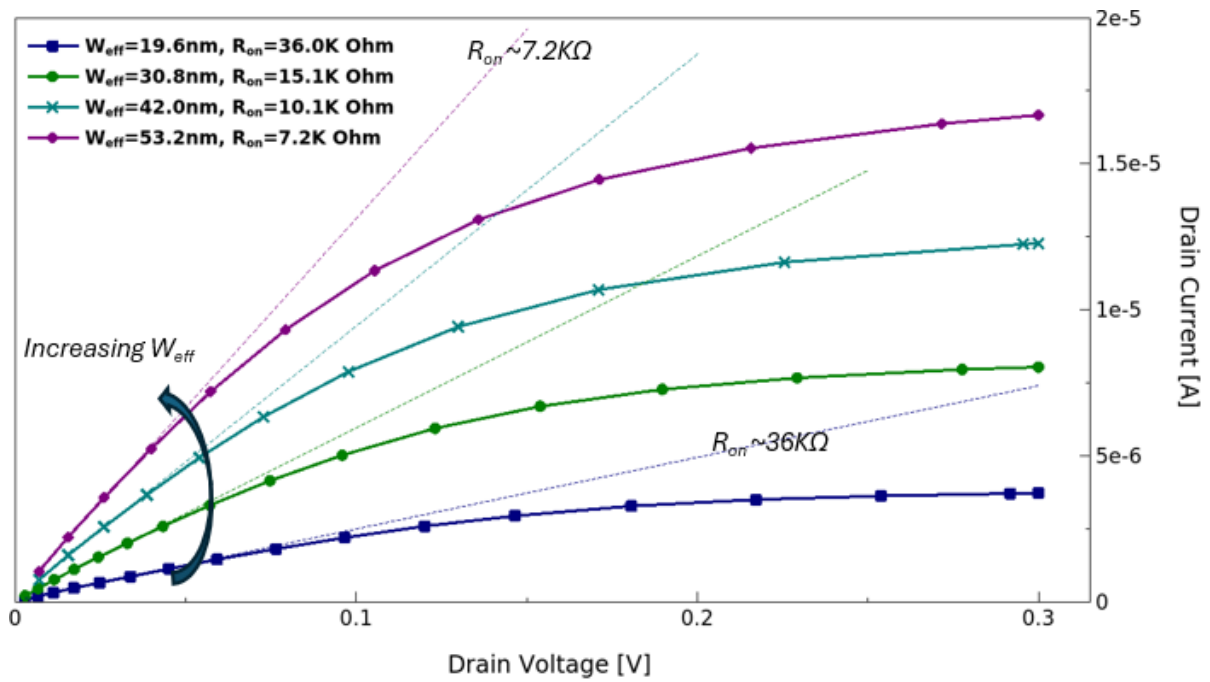


Figure 2.12 Drain voltage versus drain current across W_{eff} for $V_{gs}=0.7V$.

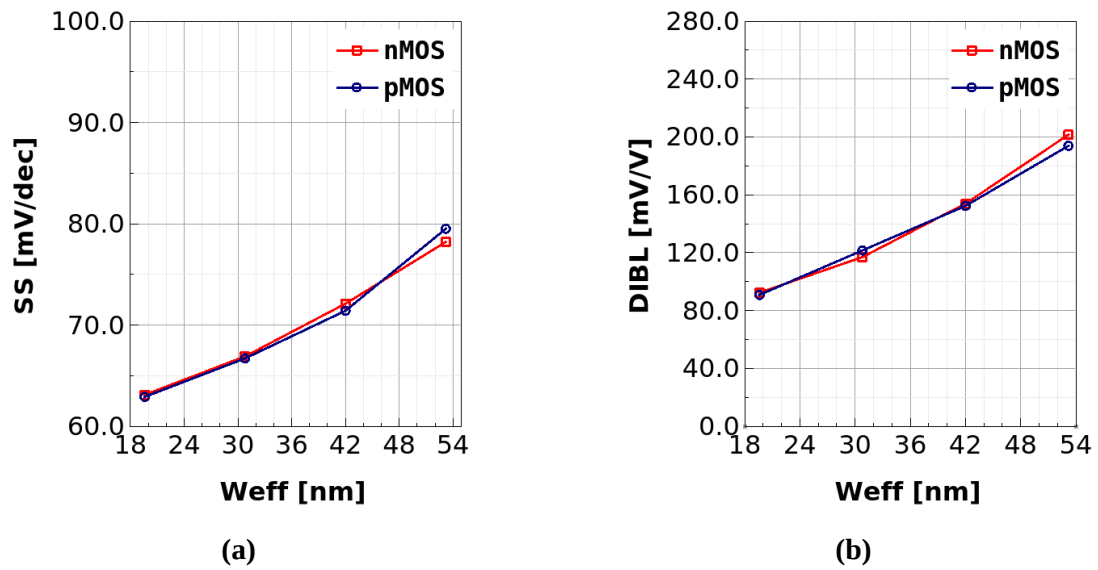


Figure 2.13 W_{eff} analysis for (a) SS and (b) DIBL

Channel Height to Width Ratio

The height to width (H/W) ratio is now varied while maintaining a constant W_{eff} to understand H/W relationship impact on device performance. Figure 2.14 shows the progression from small ratio value (i.e., small height and large width), to larger ratio value (i.e., large height and small width).

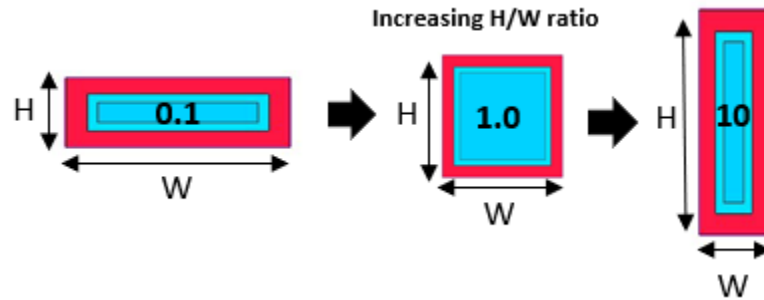


Figure 2.14 H/W ratio analysis.

The ratios at the extreme ends will produce a thin rectangular shape, while the ratios near 1.0 will produce a square shape. Good symmetry between the extreme ratios indicate there is little difference as to the orientation of a thin rectangular channel from purely a geometry standpoint, but that the thinner the channel the less I_{on} current is achievable and that subsequently less I_{off} current is incurred. A square shaped channel will enable a larger I_{on} , however, the added benefit of this somewhat plateaus and the drawback of increased I_{off} outweighs the benefits significantly. Figure 2.15a shows the mostly flat I_{on} response across geometry, with a slight peak for a square channel ratio of 1. Similarly, to the previous W_{eff} observations there is a significant drop in I_{on} current for extremely thin channels. Confirmed by I_{off} results in Figure 2.15b, this suggests too small of channel dimensions will encounter detrimental resistivity and mobility degradation from scattering and quantum confinement effects. The $I_{\text{on}}/I_{\text{off}}$ ratio is included in Figure 2.15c.

This result suggests that the best channel geometry is a rectangular shape, with channel H/W ratio in the range of 0.2 to 0.4 for best balance of I_{on} and I_{off} performance. Depending on application in either low-power or high-performance electronics, these dimensions can be adjusted according to the specific requirements. This result is in good agreement with how IRDS ultimately projected 2nm node dimensions on both low-power and high-performance devices.

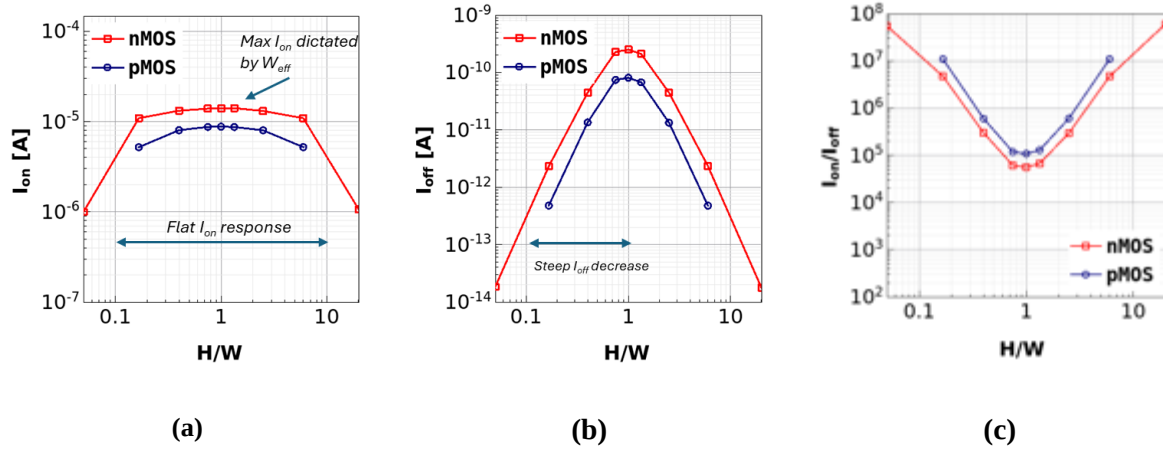


Figure 2.15 H/W analysis for (a) I_{on} , (b) I_{off} , and (c) I_{on}/I_{off} ratio.

Dual Vertically Stacked Channels

Stacking an additional channel provides a means for increasing I_{on} current while maintaining a strict device footprint. The $I_{ds}-V_{gs}$ curve in Figure 2.16 shows a near 2x increase in I_{on} current from adding a second channel. The main tradeoff to increasing I_{on} with this method is a proportional increase in I_{off} , seen by the subthreshold region of Figure 2.16.

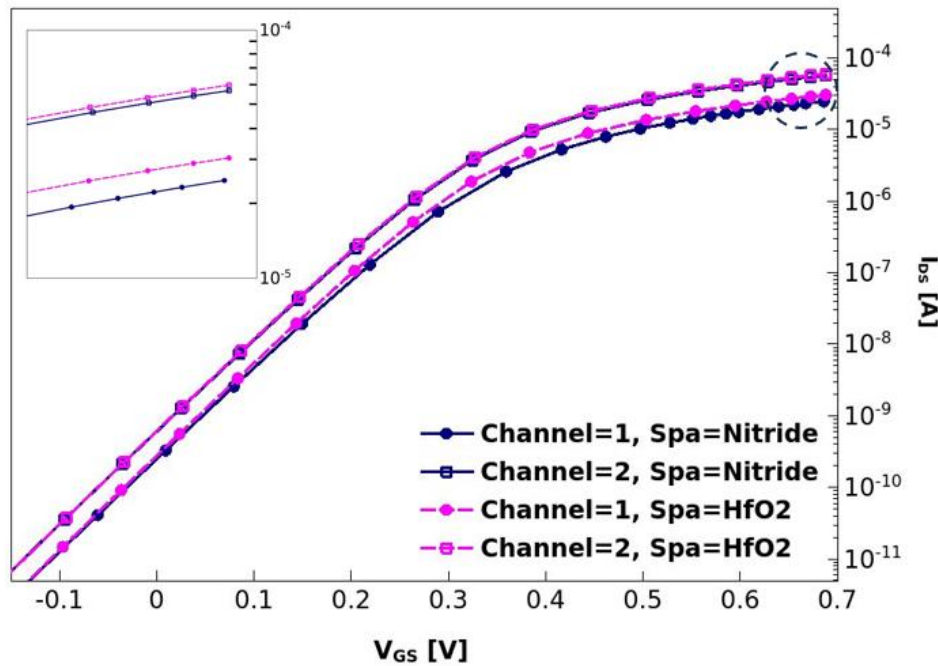


Figure 2.16 Gate voltage versus drain current comparison using low-k (Nitride) and high-k (HfO2) spacer materials on both single and dual stacked channel.

Using a high-k material for the spacer region between gate and S/D also shows additional gains in I_{on} current, indicating a boost in gate to channel and gate to channel extension region coupling. The HfO₂ spacer material appears to add additional I_{on} increase with minimal impact on I_{off} , showing similar SS as Nitride. However, an important consideration is the gate capacitance increase from using these options. Figure 2.17 shows the total gate capacitance increase for single versus dual channel, as well as for Nitride versus HfO₂ spacer material. The dual channel and HfO₂ options add significant increase to gate capacitance. Higher gate capacitance will affect the dynamic power consumption as indicated by Equation 2.4, so higher gate capacitance is necessary to consider as a tradeoff to increasing device drive with higher I_{on} current. Similarly, if gate capacitance increases too steeply, it can decrease the desired speed benefit of higher drive, since subsequent logic cells will create a larger load capacitance.

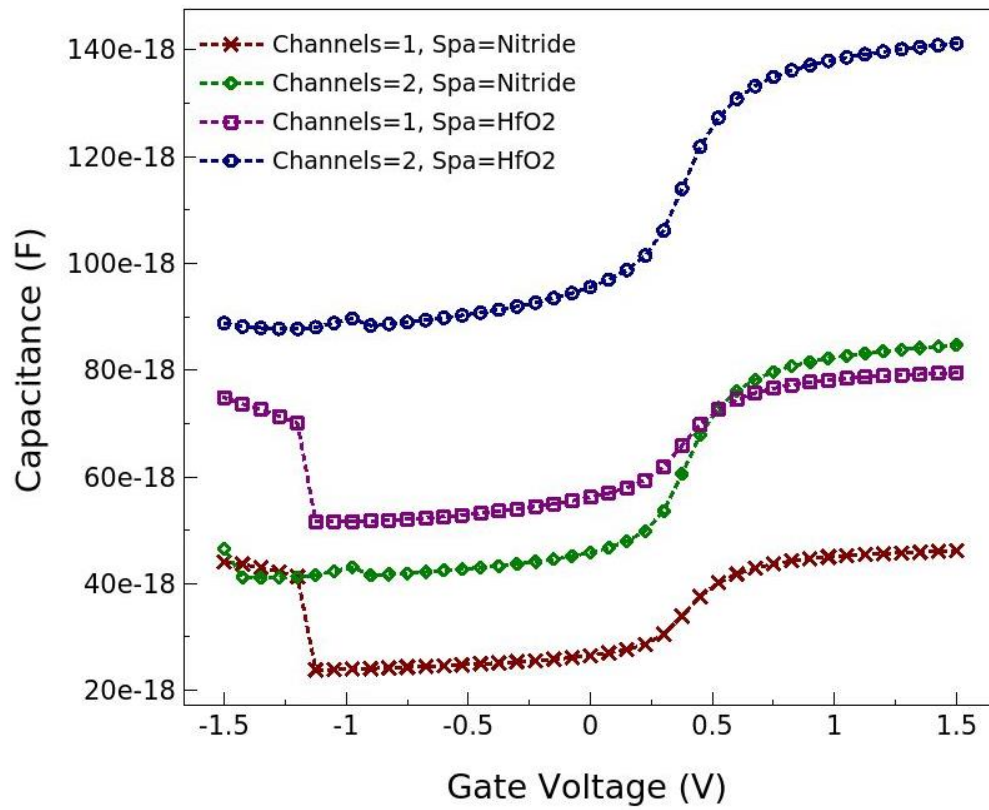


Figure 2.17 Gate capacitance for Nitride and HfO₂ spacer materials on single and dual channel devices

Chapter 3: GAAFET Circuit Performance

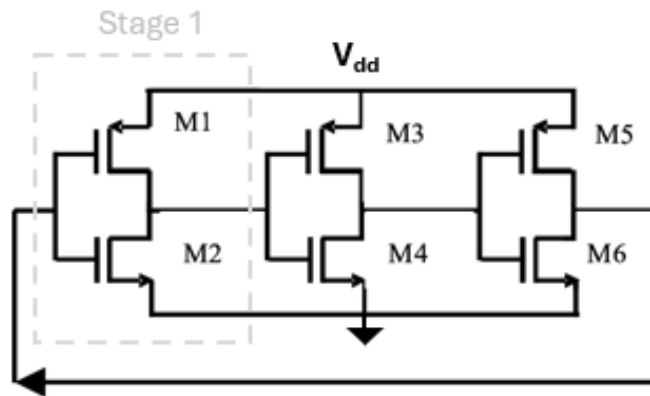
TCAD Mixed Mode circuit simulation allows for creating netlists of custom designed devices to enable quick feedback on device operation from the circuit perspective. With the growing need for Design Technology Co-optimization (DTCO) this is a powerful tool for assessing relevant circuit-based figures of merit sooner in the device design process.

Ring Oscillator (RO)

Ring oscillators are commonly used to evaluate process speed. The odd number of inverters in a closed loop introduce positive feedback that intentionally causes unstable behavior. The schematic for a 3-stage RO is shown in Figure 3.1a, as well as the first stage inverter input and output waveforms in Figure 3.1b. The frequency of oscillation can be calculated using Equation 3.1.

$$f_{osc} = \frac{1}{n \times t_{pd}} \quad (3.1)$$

where n is the number of inverters and t_{pd} is the propagation delay of both High-to-Low and Low-to-High transitions, $t_{PHL} + t_{PLH}$. A simple fan-out of 1 (F01) is used for 3-stage RO resulting in approximately a 15ps oscillation period or oscillation frequency of ~22GHz. The fanout describes the number of parallel devices (or equivalent device sizing) at each stage. Larger fanouts (e.g., F03 or F04) can be used to increase the speed of the RO and can help represent more typical loading conditions so that extracted simulation power and frequency results reflect real world expectation.



(a)

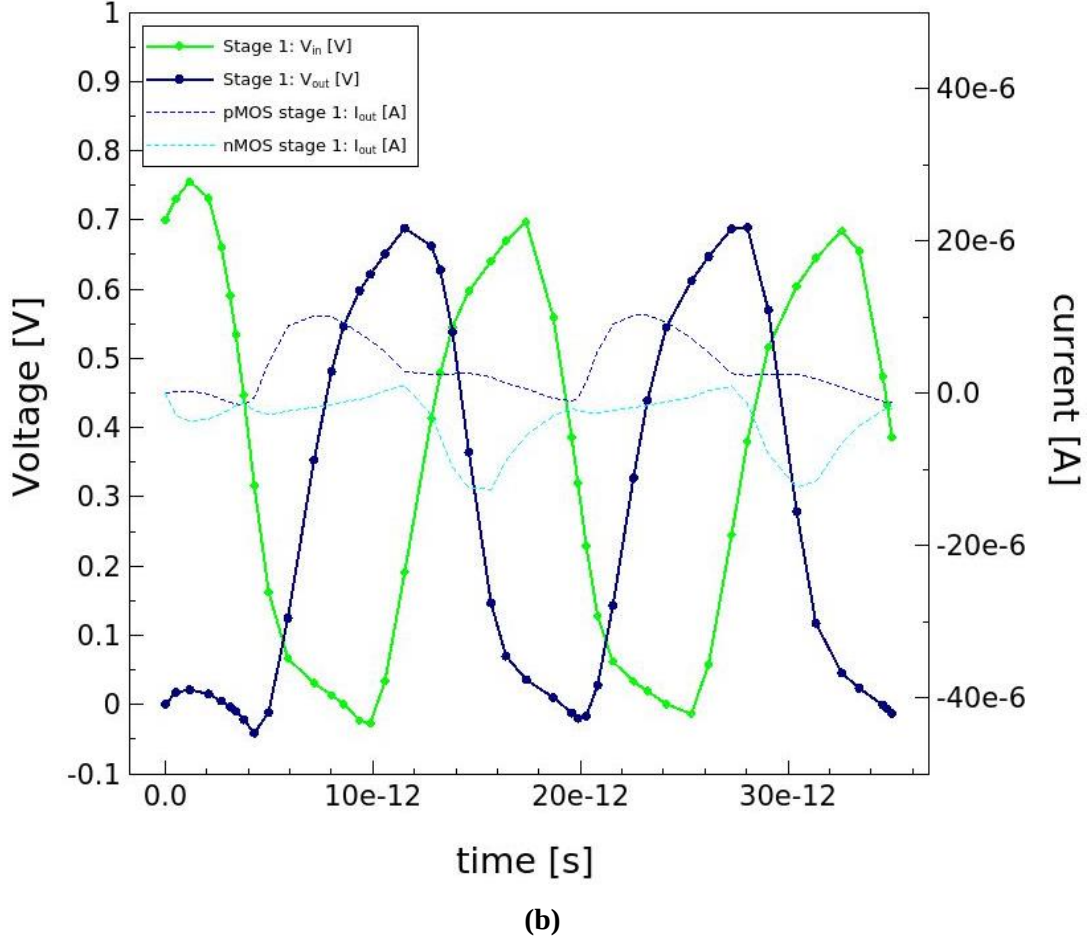


Figure 3.1 Ring Oscillator (a) schematic and (b) FO1 input/output waveforms.

Inverter

Taking a deeper dive into the RO performance means looking at the individual inverter stage performance by itself. The inverter is one of the most common CMOS circuits used in digital chips, so in depth characterization of this circuit has meaningful impact on overall chip performance. The schematic for an inverter is shown in Figure 3.2. The circuit is built by using a pMOS pull-up (PU) and nMOS pull-down (PD) network. The input signal V_{in} is connected to the gate of both PU and PD networks. The PU network connects to the voltage supply rail, while the PD network connects to the reference or ground rail. The circuit output, V_{out} , is then connected to either the supply rail when V_{in} is low, creating a logic one output, or to ground when V_{in} is high, creating a logic zero output. From the C-V evaluation, an 80aF load is chosen for C_{out} to replicate the realistic load capacitance of driving another inverter stage. The V_{in} and V_{dd} voltages are simply modeled using an ideal voltage source.

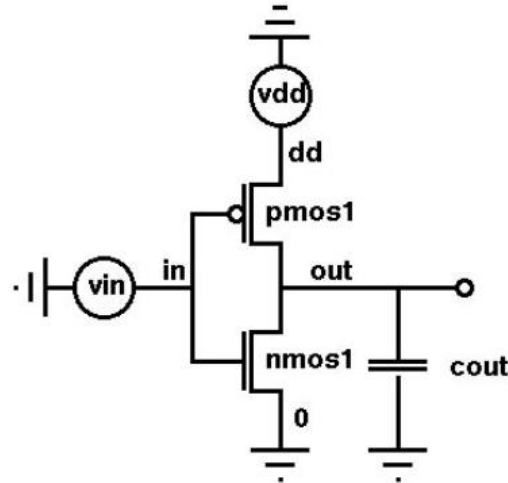


Figure 3.2 Inverter schematic [Sentaurus TCAD manuals, 2020].

Simulating this environment, we can evaluate the voltage transfer curve (VTC) for when V_{in} switches from low to high, shown by Figure 3.3. The switching point, V_{sw} , is where $V_{in} = V_{out}$, on the VTC curve. The ideal switching point, marked by the diagonal line, intersects this point exactly at $V_{dd}/2$ [Weste, Neil H. E., Harris, David Money, 2015]. The switching current is overlayed to show the difference between $P_{dynamic}$ and P_{static} current profiles. $P_{dynamic}$ is given in Equation 2.4 and P_{static} is determined by $I_{off} \times V_{dd}$. Although I_{off} appears minimal compared to I_{on} , this has continued to be an increasing concern due to the growing number of transistors per semiconductor chip. Also, because P_{static} does not contribute to any meaningful computation, this is strong motivation for maintaining low I_{off} on a device level basis.

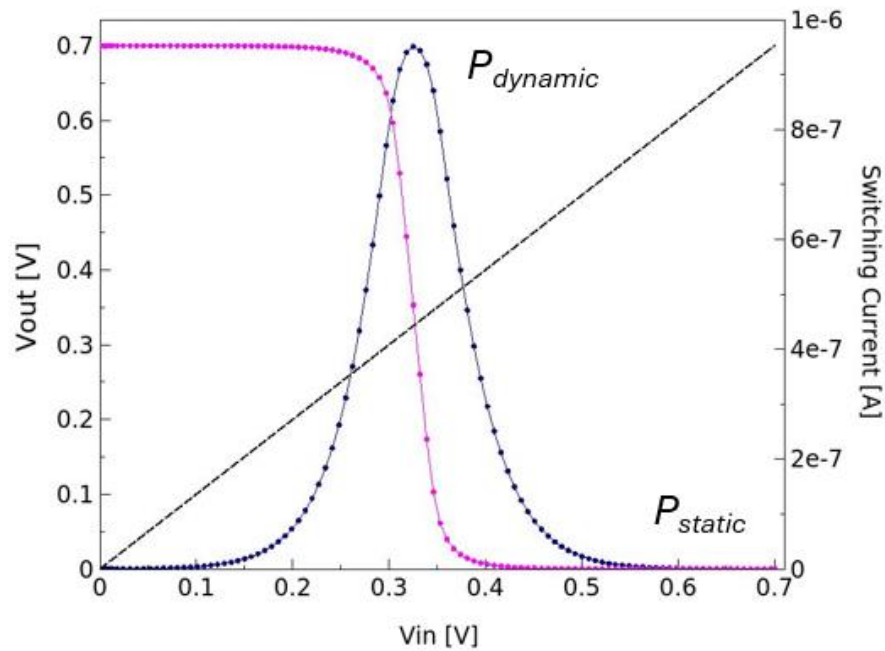


Figure 3.3 Inverter voltage transfer curve (VTC).

A transient simulation in depicted Figure 3.4 shows the ideal voltage source V_{in} signal with 25ps rise and fall time, and the subsequent inverter V_{out} signal. The V_{out} signal begins to show some characteristic RC delay from charging the 80aF load through the on-state resistance of the PU and PD devices. Overlayed on this graph is the alternating current flowing into and out of the load capacitance. Notably, a much larger current is seen during the initial switch that will discharge the capacitor from its fully charged state before a more symmetrical charge/discharge cycle is seen. Current alternates between positive and negative values as the inverter switches from either sinking current when the nMOS is active and supplying current when the pMOS is active. The current polarity is reflected by this since it is plotted from the C_{out} perspective. Similar to the VTC in Figure 3.3, the majority of current is seen only during switching operations.

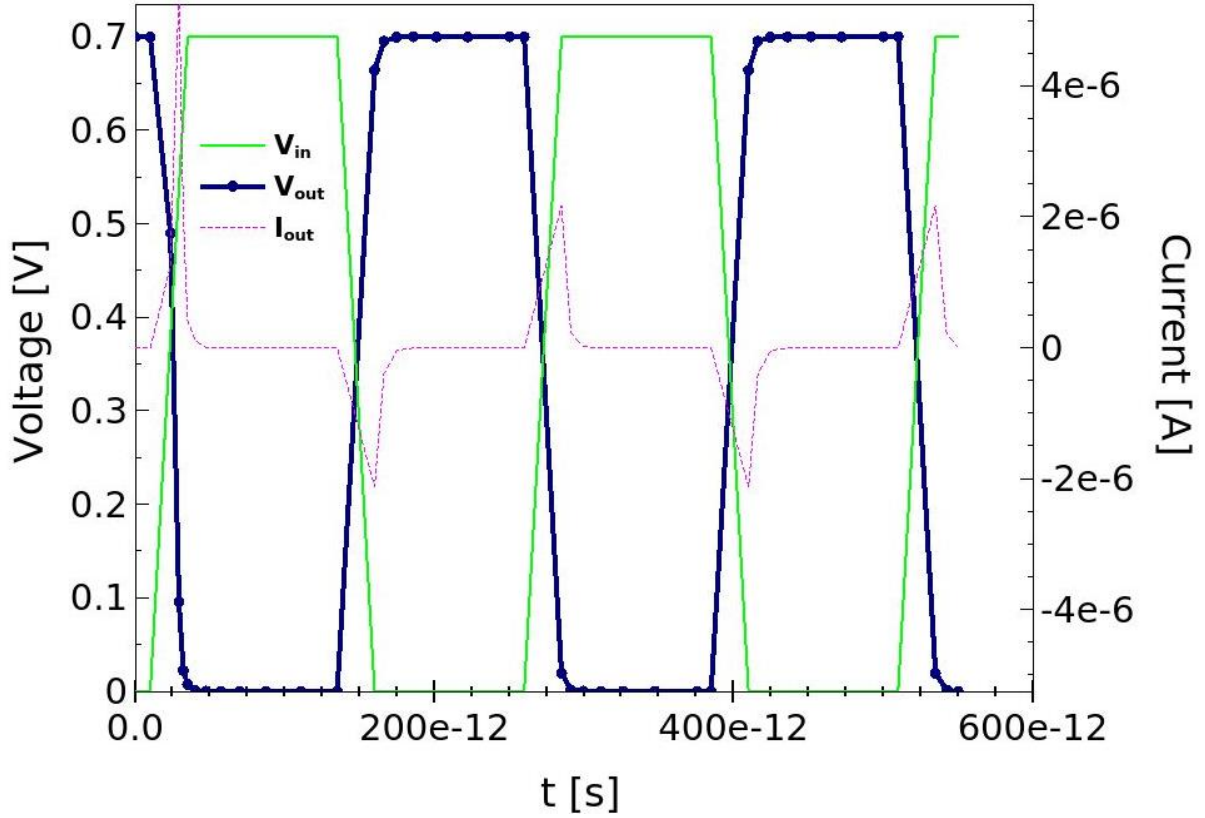


Figure 3.4 Inverter transient simulation.

Noise Margins

An ideal VTC occurs when the PU and PD devices are well matched, so that the switching point is at or very close to $V_{dd}/2$. This gives the inverter equal noise margin for both logical 1 and logical 0 inputs. The VTC in Figure 3.5a was generated using an nMOS and pMOS with identical device widths. Plotting the ideal V_{sw} , it is apparent that the inverter has asymmetry in its transfer characteristics. The switching point is approximately 325mV (instead of $V_{dd}/2 = 350\text{mV}$). This translates as a logical 1 having more noise margin than a logical 0. Noise margins are calculated using Equation 3.2.

$$NM_L = V_{IL} - V_{OL} \quad (3.2)$$

$$NM_H = V_{OH} - V_{IH}$$

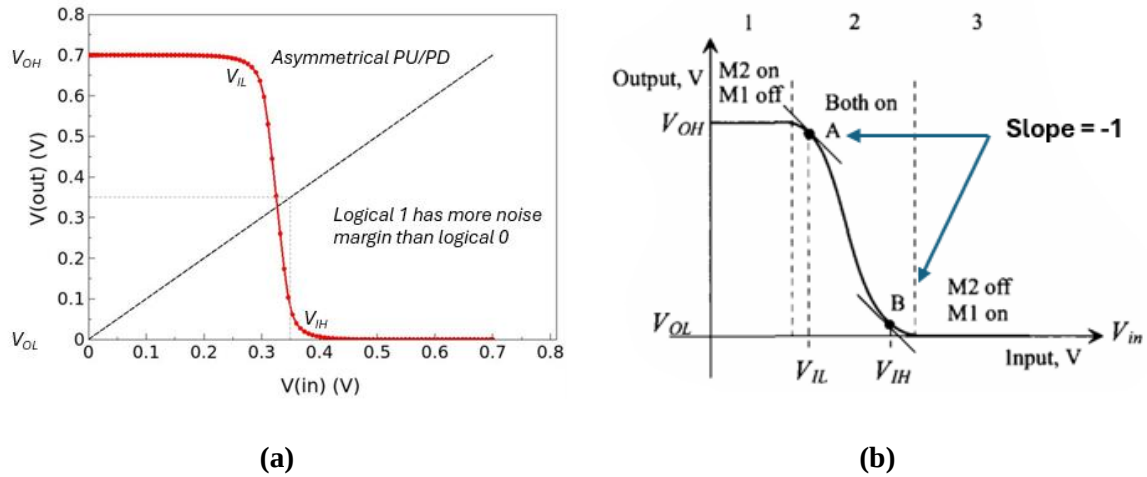


Figure 3.5 Inverter (a) asymmetrical VTC and (b) V_{IH} and V_{IL} [Baker, 2019]

The V_{IL} and V_{IH} points, A and B in Figure 3.5b, can be calculated by taking the first order derivative of the VTC curve and recording the V_{in} where the slope is -1. Any input values less than V_{IL} will output a full logic one equal to V_{OH} , while any input values greater than V_{IH} will output a logic zero equal to V_{OL} . Input values greater than V_{IL} but less than V_{IH} can lead to metastability, meaning that the output can result in either a logic one or logic zero and may take an undetermined amount of time to resolve one way or the other. Larger device gain creates a steeper VTC curve, bringing V_{IL} and V_{IH} closer together, reducing the risk of metastability occurring.

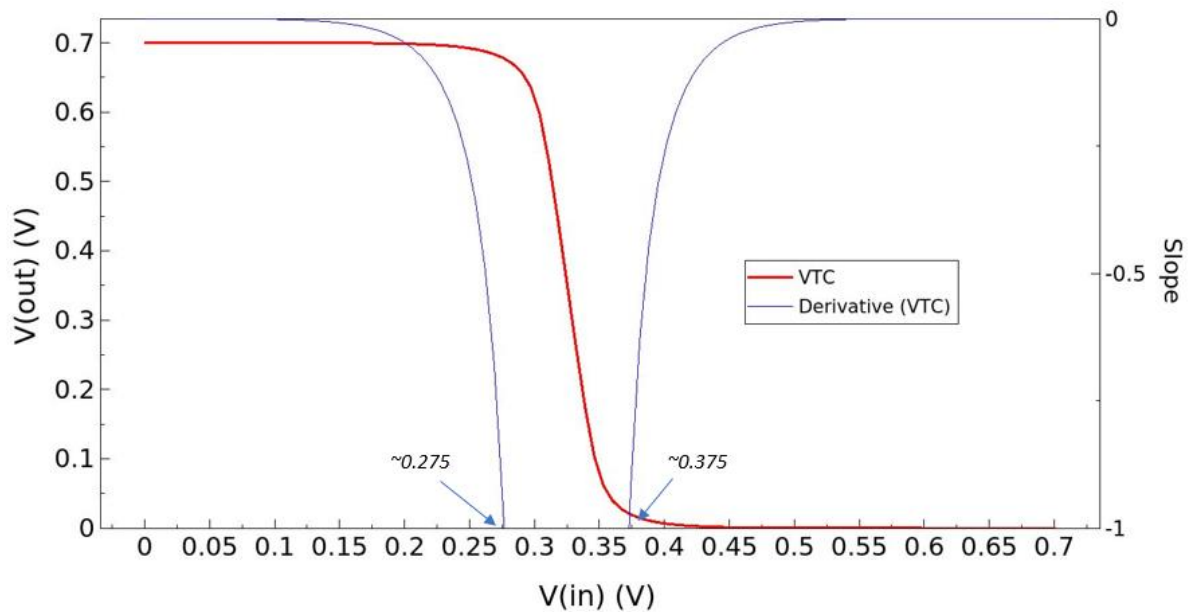


Figure 3.6 First order derivative of the asymmetrical inverter VTC.

Table 3.1 Asymmetrical inverter noise margins

NM_H	325mV
NM_L	275mV

The derivative of the VTC curve is shown graphically in Figure 3.6. Plugging these values into Equation 3.2, the noise margin for logic 0 and logic 1 are calculated and results shown in table 3.1. This describes the amount of input fluctuation from a noisy environment (i.e., the input logic levels are not perfectly at V_{dd} or ground) the inverter can tolerate before potentially risking a metastable state. To provide more symmetric noise margins for the PU and PD networks, the widths of the devices can be adjusted so that the resistance, and resulting drives, are equal. The result of increasing the pMOS width shows how matched inverter drive moves the V_{sw} to the $V_{dd}/2$ point, shown in Figure 3.7.

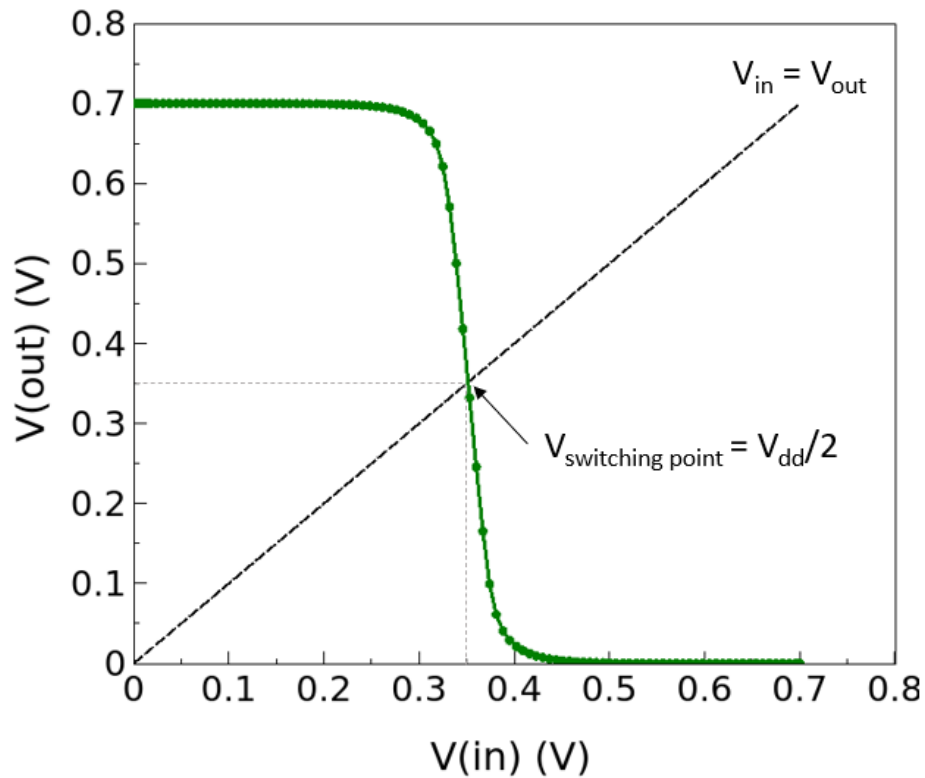


Figure 3.7 Matched inverter PU/PD network VTC.

Chapter 4: Conclusion

This thesis explores some of the fundamental MOSFET device characterization steps to better understand impacts of different design choices. As semiconductor scaling continues to shrink, the margin for error shrinks with it. This emphasizes the need for strong understanding of the potential roadblocks that SCE may introduce for MOSFETs on the nanometer scale. As shown in this work, leveraging TCAD is a powerful way to explore options and accurately assess device characteristics. Taking a top-down approach, many aspects of the GAAFET structure are analyzed to determine the best parameter options. Starting with channel effective width and geometry, physically accurate models help to create a framework for best implementation. Results from sweeping channel dimensions align well with the height and width specifications projected by IRDS. This showed H/W ratios in range of 0.2 and 0.4 will provide the best tradeoffs between I_{on}/I_{off} ratio. These results should be interpreted as guidelines and indicate parameters like W_{eff} can be modulated in conjunction with H/W ratio to meet end application needs. While the GAAFET structure is not immune to SCE, the increased electrostatic control of the channel can help significantly reduce their impact. Tolerable SCE, like SS and DIBL, can be achieved with the GAAFET structure, underlining that channel geometry engineering, vertically stacked channels, and material choices are useful knobs to adjust performance. Parasitic extraction is necessary to understand tradeoffs to these types of choices, shown by fundamental methodologies for resistance and gate capacitance parameter extraction that are analyzed and discussed. Circuit level performance metrics, such as RO and inverter VTC, are simulated to provide additional context for device level results.

Ultimately, the transition to GAAFET architecture provides an optimistic outlook that with the right tailored approach, GAAFET can enable viable solutions for the 2nm process node (and possibly beyond).

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